DISCRETIO - What Do World Models *Implicitly Learn*? Revealing Latent Behavioral Priors Through Future Rollouts

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Abstract

World models are increasingly positioned as the **predictive substrate of embodied intelligence**, enabling simulation, planning, and action under uncertainty. Yet current evaluations largely ask whether these models can *predict future frames*, *forecast trajectories*, reconstruct hidden observations, or improve downstream control. These are necessary tests, but they miss a deeper question: **what do world models implicitly learn beyond visible observation?**

We introduce DISCRETIO, a benchmark for revealing **latent behavioral priors** in predictive world models through their *future rollouts*. Rather than treating prediction accuracy as the sole endpoint, DISCRETIO probes the hidden assumptions that shape prediction when evidence is incomplete, occluded, ambiguous, or socially contextualized. For example, when a cyclist disappears behind a bus, a model may continue to reserve future occupancy for the cyclist, slow the predicted motion, or incorrectly let the agent vanish; similarly, when an adult, child, and elderly pedestrian approach the same crossing with matched position and velocity, differences in predicted stopping, crossing, or uncertainty reveal actor-specific priors. These priors are not assumed to be symbolic; they are measured through systematic shifts in predicted trajectories, future occupancy, uncertainty dispersion, collision margins, and multimodal rollout structure.

DISCRETIO contains four controlled task families. **Physical-belief** tasks test object permanence and hidden-state tracking under occlusion: a cyclist enters behind a bus with known speed and heading, and the model must predict whether the cyclist continues, slows, turns, or vanishes from future occupancy. **Behavioral-belief** tasks test actor-specific expectations: an adult, child, and elderly pedestrian approach the same crossing with matched position, speed, and heading, and the benchmark measures whether predicted uncertainty, crossing probability, or stopping probability changes despite identical observed motion. **Social-belief** tasks test role-based interaction priors: two agents approach the same narrow passage under identical geometry, but one is marked as a delivery rider and the other as an ordinary commuter, revealing whether role identity alone shifts predicted yielding, acceleration, or collision avoidance. **Counterfactual-belief** tasks test contextual priors under fixed dynamics: the same crossing trajectory is rendered in a structured U.S. crosswalk, a dense Indian market street, and an unmarked road, exposing cultural or environmental priors beyond observed motion.

We evaluate latent predictive world models, generative video world models, trajectory forecasting systems, and embodied planning models using multi-sample future rollouts. DISCRETIO reports metrics for **prior persistence**, **belief revision**, **false-belief consistency**, **counterfactual sensitivity**, **calibration**, and **prior amplification** relative to empirical uncertainty. By reframing world-model evaluation from *prediction quality* to *prior discovery*, DISCRETIO enables systematic auditing of the hidden assumptions that govern future simulation, providing a foundation for more trustworthy, interpretable, and socially aware embodied AI.

1 What Do World Models Learn Beyond Observation?

World models are becoming the **predictive substrate of embodied intelligence**: they simulate future states, support planning, and guide action under uncertainty (Ha and Schmidhuber, 2018; Hafner et al., 2020, 2023; LeCun, 2022). Yet most evaluations still ask a narrower question: *can the model predict what happens next?* This question is necessary, but incomplete. A model may forecast plausible futures while relying on hidden assumptions about **persistence**, **intent**, **social roles**, **risk**, or **context**. Thus, the central question is not only whether a world model predicts, but **what it implicitly learns beyond direct observation**.

1.1 World Models as Predictive Systems

The defining property of a world model is its ability to anticipate states that are not currently observed. Early systems learned compact latent dynamics for imagination-based control (Ha and Schmidhuber, 2018; Hafner et al., 2020); later approaches scaled this idea to diverse control domains (Hafner et al., 2023). More recent predictive architectures, including JEPA-style models, emphasize learning representations that support prediction without necessarily reconstructing pixels (LeCun, 2022; Assran et al., 2025). Across these paradigms, the objective is shared: **learn a representation of the world that can be rolled forward**.

However, *rolling the world forward* requires more than extrapolating visible motion. When objects are occluded, agents hesitate, roles differ, or environments change, the model must fill gaps using learned priors. These priors are often invisible to standard accuracy metrics.

1.2 Prediction Is Not Understanding

Current world-model evaluations primarily measure *prediction quality*: future-frame accuracy, trajectory error, reconstruction fidelity, planning success, or downstream control performance. Such metrics are valuable, but they evaluate the *output* of prediction, not the assumptions that produced it.

This distinction matters. Two models may achieve similar trajectory error while encoding different latent expectations. One model may preserve an occluded cyclist as a continuing hidden entity; another may let the cyclist disappear once no longer visible. One model may assign higher hesitation probability to a child at a crossing; another may treat all pedestrians identically when position and velocity are matched. These differences reflect **latent behavioral priors**: learned expectations that shape future rollouts when evidence is *incomplete*, *ambiguous*, or *socially contextualized*.

Thus, **prediction accuracy is not enough**. It tells us whether a future matched the data, but not *why the model imagined that future*.

1.3 What Existing Benchmarks Measure

Existing benchmarks have substantially advanced world-model evaluation. Physical reasoning benchmarks test whether models respect object permanence, contact, support, and physical consistency (**???**). Video prediction and forecasting benchmarks measure whether models generate plausible future observations. Planning and control benchmarks evaluate whether learned dynamics improve downstream decision making (Hafner et al., 2020, 2023). Representation-learning benchmarks test whether predictive features support recognition, forecasting, or action (LeCun, 2022; Assran et al., 2025).

Evaluation Focus	Performance	Prior Discovery	Representative Work
Video prediction / world modeling	✓	✗	Future rollouts and latent imagination (Ha and Schmidhuber, 2018; Hafner et al., 2020, 2023)
Physical reasoning	✓	✗	Physical plausibility, permanence, continuity, solidity (**???**)
Planning / control	✓	✗	Downstream decision making through learned dynamics (Hafner et al., 2020, 2023)
Predictive representation learning	✓	✗	Predictive representations for understanding, prediction, and planning (LeCun, 2022; Assran et al., 2025)
DISCRETIO	✓	✓	**Latent behavioral priors behind future rollouts**

Table 1: **Evaluation gap.** Existing evaluations measure whether predictive systems perform well, but do not systematically reveal the *latent behavioral priors* that shape their predicted futures. DISCRETIO explicitly targets this missing dimension.

The missing dimension is therefore not prediction, physical consistency, representation quality, or control. The missing dimension is **prior discovery**: identifying the hidden assumptions that govern future simulation.

1.4 Hidden Assumptions in Future Rollouts

Future rollouts make latent priors observable. When evidence is complete, a model may simply follow visible dynamics. But when evidence becomes incomplete, occluded, ambiguous, or context-dependent, the model must rely on learned expectations.

For example, if a cyclist moves behind a bus with known speed and heading, the model must decide whether the cyclist *continues*, *slows*, *turns*, *stops*, or *vanishes* from future occupancy. If an adult, child, and elderly pedestrian approach the same crossing with identical position, speed, and heading, differences in predicted stopping, crossing, or hesitation reveal **actor-specific behavioral priors**. If a delivery rider and an ordinary commuter enter the same narrow passage under identical geometry, changes in yielding or collision avoidance reveal **role-based social priors**. If the same crossing trajectory is rendered in a structured U.S. crosswalk, a dense Indian market street, and an unmarked road, shifts in prediction reveal **contextual priors beyond observed motion**.

These priors need not be symbolic. They may appear as changes in *trajectory distributions*, *future occupancy*, *uncertainty dispersion*, *collision margins*, or *multimodal rollout structure*. The key insight is simple: **what a model rolls out reveals what it has implicitly learned**.

1.5 The Missing Evaluation Dimension

We argue that world-model evaluation requires a shift from *prediction quality* to *prior discovery*. Existing benchmarks ask: *Did the model predict the future correctly?* We ask: **What assumptions shaped the predicted future?**

To formalize this question, we organize latent behavioral priors into four families:

- **Physical Priors**: object permanence, hidden-state tracking, and motion continuity.
- **Behavioral Priors**: intent inference, hesitation expectation, and risk sensitivity.
- **Social Priors**: role priors, yielding expectations, and interaction norms.
- **Contextual Priors**: cultural priors, environmental priors, and contextual expectations.

These priors capture what world models learn beyond visible observation. They determine whether hidden entities persist, whether agents are expected to hesitate, whether roles alter interaction dynamics, and whether context changes imagined futures.

To our knowledge, no existing benchmark systematically measures latent behavioral priors in predictive world models under controlled variations of hidden state, agent identity, social role, and environmental context. We introduce DISCRETIO to fill this gap: a benchmark for revealing, quantifying, and comparing the priors that shape future rollouts.

2 Latent Behavioral Priors in Predictive World Models

Section 1 showed that prediction quality alone does not reveal what a world model has implicitly learned. We now formalize the missing object of evaluation: **latent behavioral priors**. These priors are not hand-coded rules, symbolic beliefs, or explicit world states. They are *learned expectations* that shape future rollouts whenever observation alone is insufficient to determine what happens next.

A latent prior is not what a model predicts; it is what makes one future more likely than another when multiple futures remain compatible with the same evidence. This distinction is central. Prediction tells us the model's output; priors reveal the assumptions that govern that output.

2.1 From Prediction to Prior Formation

A world model never observes the future directly. It observes a partial history and must roll that history forward. When the evidence is complete, prediction may largely follow visible dynamics. But when an object is occluded, an agent hesitates, a social role matters, or the same motion occurs in a different context, the model must fill gaps using learned expectations.

Definition: Latent Behavioral Prior

A **latent behavioral prior** is a learned expectation that *systematically influences future rollouts* when multiple futures remain compatible with the observed evidence.

Latent priors are therefore not directly observed. They are inferred from their consequences: shifts in *future occupancy*, *trajectory distributions*, *uncertainty dispersion*, *collision margins*, interaction timing, or *multimodal rollout structure*. In this sense, future prediction becomes a measurement instrument: **what a model rolls out reveals what it has implicitly learned**.

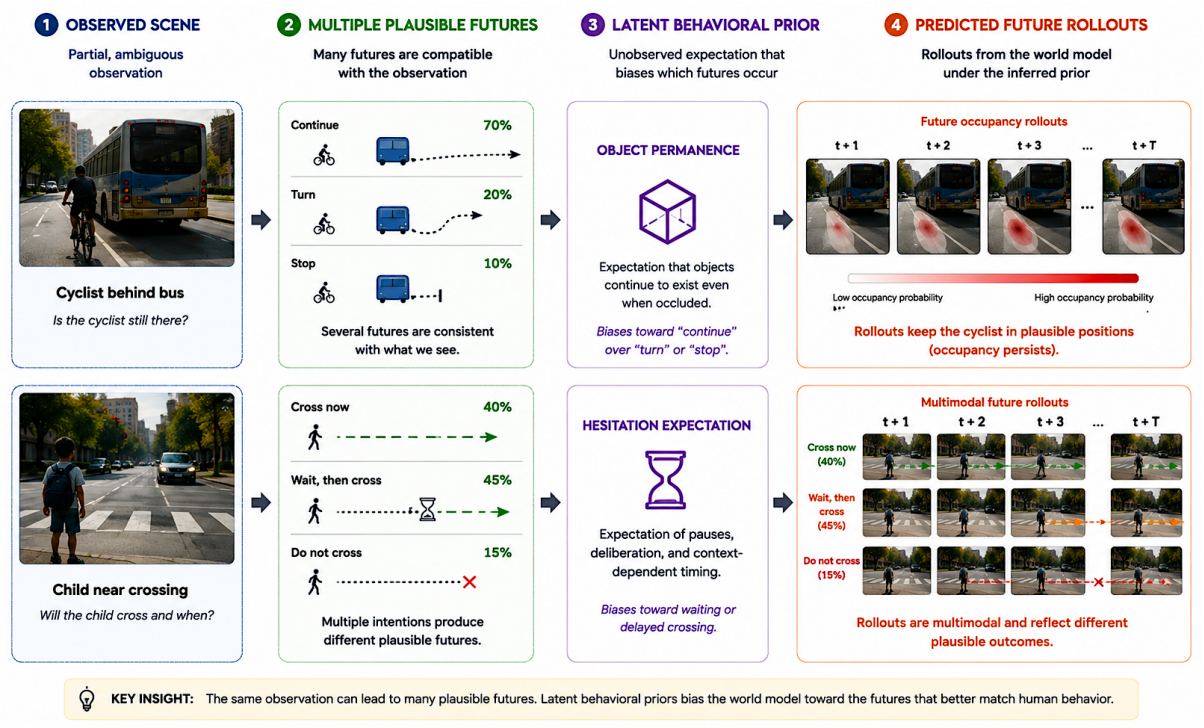


Figure 1: **From observation to prior-guided future rollouts.** A partial or ambiguous observation can admit multiple plausible futures. Latent behavioral priors bias the world model toward particular futures, and these biases become visible through rollout structure. In the physical case, *object permanence* preserves future occupancy for an occluded cyclist. In the behavioral case, *hesitation expectation* produces delayed or multimodal crossing rollouts.

2.2 Four Families of Latent Behavioral Priors

We organize latent behavioral priors into four families: **physical**, **behavioral**, **social**, and **contextual**. Each family captures a different kind of expectation that world models may acquire beyond visible observation.

Physical priors. **Physical priors** capture assumptions about entities and motion when parts of the scene are no longer visible. They answer: *what remains true when something is not currently observed?* A model should preserve object permanence when a cyclist disappears behind a bus, track hidden state when a pedestrian passes behind a truck, and maintain motion continuity when a vehicle enters a blind turn. The goal is not exact prediction under occlusion, but **coherent uncertainty**: hidden entities should remain possible, spatially plausible, and dynamically consistent.

Behavioral priors. **Behavioral priors** capture expectations about what agents are likely to do next. They answer: *what action is this agent likely to take?* A pedestrian near a curb may cross, wait, turn back, or continue walking. A child approaching a crossing may hesitate even when its observed trajectory matches an adult's. An elderly pedestrian may induce wider predicted safety margins than a young adult under identical motion. These cases reveal priors over **intent**, **hesitation**, and **risk-sensitive behavior**.

Social priors. **Social priors** capture expectations that arise from interaction among agents. They answer: *how does who an agent is change what the model expects them to do?* A delivery rider and an ordinary commuter may approach the same narrow passage with identical geometry and motion, yet induce different predictions about acceleration, yielding, or collision avoidance. Multi-agent scenes

further require assumptions about priority, cooperation, and coordination. These priors are expressed through **role-conditioned rollouts**, **yielding behavior**, and **interaction norms**.

Contextual priors. **Contextual priors** capture expectations induced by the environment itself. They answer: *how does the same observed motion change meaning across contexts?* The same crossing trajectory may produce different futures in a structured U.S. crosswalk, a dense Indian market street, an unmarked road, or a school zone. Such shifts reveal **cultural**, **environmental**, and **scene-semantic** priors beyond observed motion.

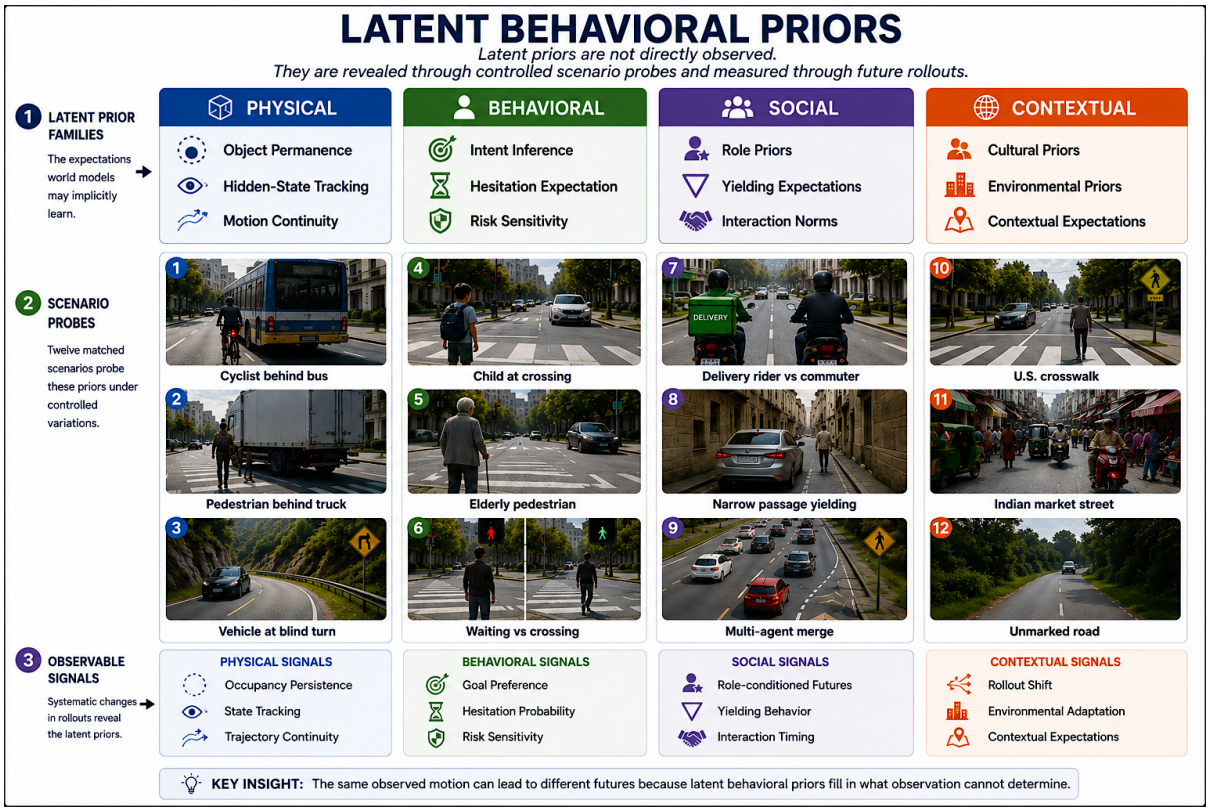


Figure 2: **Taxonomy of latent behavioral priors.** DISCRETIO organizes priors into four families: **physical**, **behavioral**, **social**, and **contextual**. Each family is exposed through controlled scenario probes and measured through *future rollout signals*, such as occupancy persistence, goal preference, yielding behavior, and context-dependent rollout shifts.

2.3 From Latent Priors to Observable Rollout Signals

Latent behavioral priors are not accessible as internal labels. They must be inferred from rollout-level evidence. A prior becomes measurable when a controlled change in hidden state, actor identity, social role, or context produces a systematic shift in predicted futures.

Table 9 links each prior family to its observable signal.

2.4 A Taxonomy for Prior Discovery

The resulting taxonomy separates the *object of measurement* from the benchmark protocol. The object of measurement is the latent prior; the protocol is the controlled future-rollout setting used to expose it. This distinction is important: DISCRETIO is not merely a collection of scenarios, but a framework for asking **which hidden assumptions govern predicted futures**.

In the next section, we operationalize this taxonomy through DISCRETIO: a benchmark that constructs matched scenarios where hidden state, agent identity, social role, or environmental context changes while visible motion is controlled. This allows us to move from *prediction accuracy* to *prior discovery*: not only asking whether a model predicted the future, but **what assumptions made that future likely**.

Prior Family	Latent Prior	Observable Rollout Signal
Physical	Object permanence	Future occupancy persists after occlusion
Physical	Hidden-state tracking	Uncertainty expands but remains spatially coherent
Physical	Motion continuity	Predicted paths respect speed, heading, and dynamics
Behavioral	Intent inference	Future paths diverge toward plausible goals
Behavioral	Hesitation expectation	Delayed motion, stopping probability, or multimodal hesitation
Behavioral	Risk sensitivity	Wider safety margins or reduced crossing confidence
Social	Role priors	Role-conditioned shifts in acceleration, yielding, or avoidance
Social	Yielding expectations	Changes in priority, timing, and collision margins
Social	Interaction norms	Coordinated multi-agent rollouts rather than independent motion
Contextual	Cultural priors	Different futures under matched motion across cultural settings
Contextual	Environmental priors	Scene-density or road-structure dependent prediction shifts
Contextual	Contextual expectations	Scene semantics change predicted intent, timing, or uncertainty

Table 2: **From priors to observable signals.** Latent behavioral priors are inferred from systematic changes in future occupancy, trajectories, uncertainty, timing, collision margins, and interaction structure.

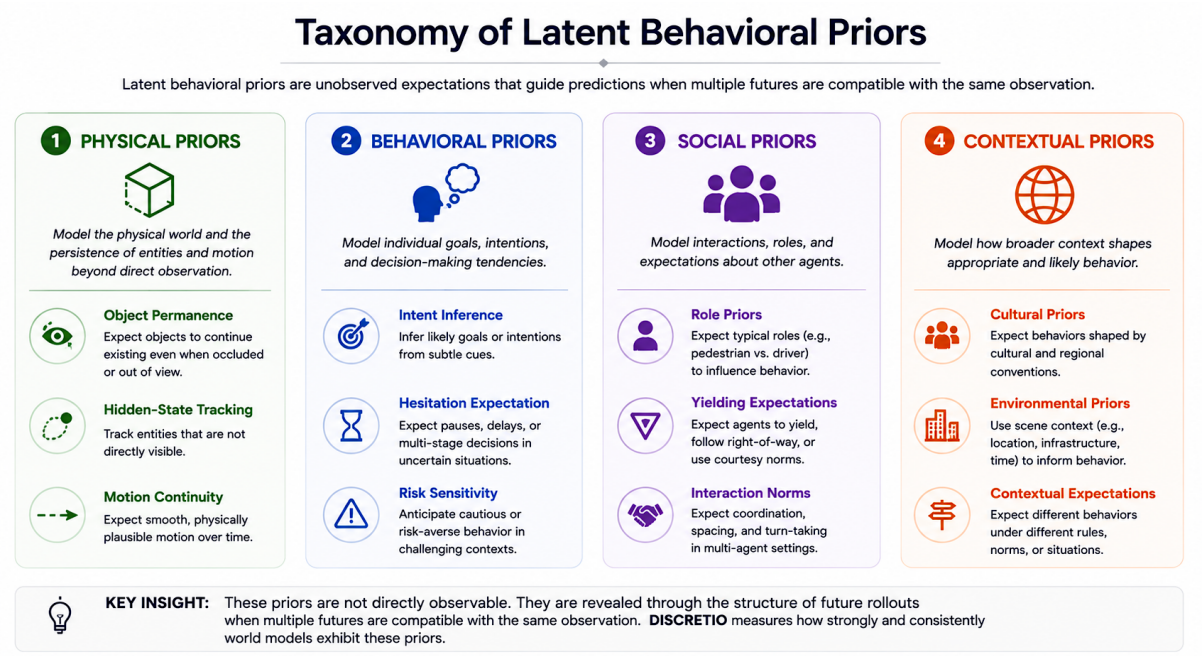


Figure 3: **Taxonomy of latent behavioral priors.** We organize the implicit expectations of predictive world models into four families: **physical**, **behavioral**, **social**, and **contextual** priors. These priors are not directly observable; DISCRETIO reveals them through controlled future rollouts when multiple futures are compatible with the same observation.

3 DISCRETIO: A Benchmark for Measuring Latent Priors

The central premise of DISCRETIO is that latent behavioral priors cannot be measured through prediction accuracy alone. Instead, they must be revealed through controlled future-rollout probes that isolate hidden assumptions while preserving observable evidence. To achieve this, we construct a benchmark from approximately 700 hours of real-world Indian suburban video and systematically

transform naturally occurring situations into controlled prior-discovery scenarios.

Unlike conventional forecasting benchmarks, which evaluate whether a model predicts the future correctly, DISCRETIO evaluates how future predictions change when hidden state, agent identity, social role, or environmental context is manipulated under controlled conditions. The benchmark therefore treats future rollouts as a measurement instrument for exposing latent behavioral priors.

3.1 Benchmark Overview

DISCRETIO is constructed through a multi-stage pipeline that converts raw video into controlled benchmark probes.

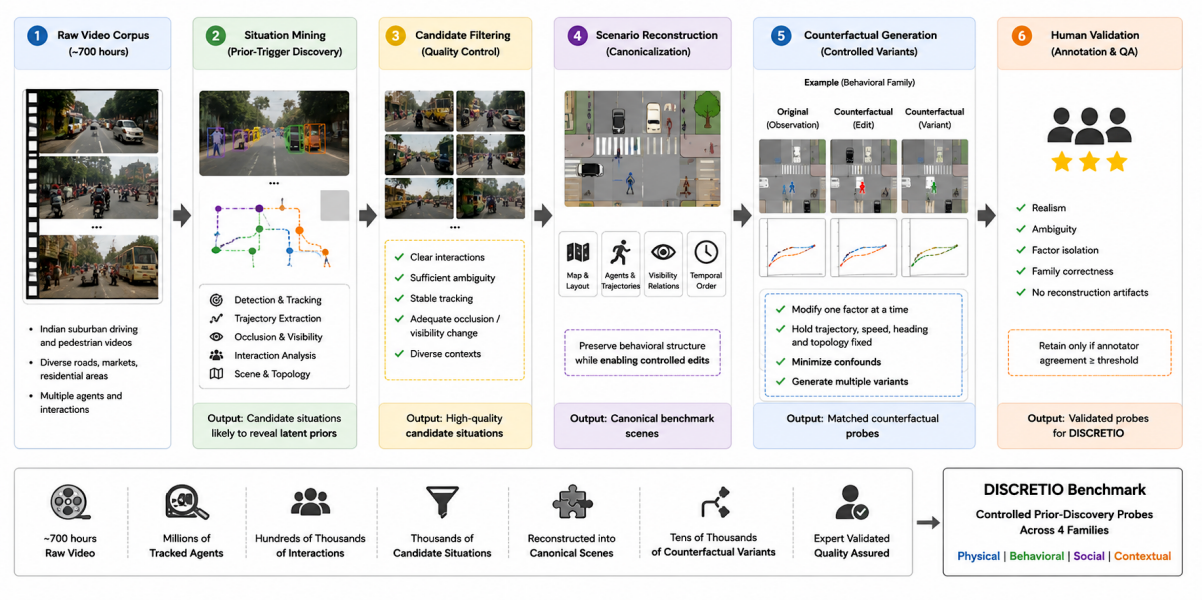


Figure 4: **DISCRETIO construction pipeline.** Raw video is transformed into controlled prior-discovery probes through situation mining, candidate filtering, scenario reconstruction, counterfactual generation, and human validation.

The pipeline consists of five stages:

1. Mining naturally occurring prior-triggering situations.
2. Filtering and quality control.
3. Reconstruction into canonical benchmark scenes.
4. Controlled counterfactual generation.
5. Human validation and benchmark assembly.

While DISCRETIO is constructed from Indian suburban video, the benchmark does not evaluate geographic knowledge. Rather, real-world video serves as a source of naturally occurring ambiguity, occlusion, interaction, and contextual variation from which controlled prior-discovery probes are derived.

3.2 Mining Prior-Triggering Situations

The source corpus contains approximately 700 hours of suburban driving and pedestrian video, covering dense traffic, informal crossings, roadside activity, markets, residential streets, and mixed-agent interactions.

Videos are segmented into overlapping temporal windows and processed using object detection, multi-object tracking, trajectory estimation, scene parsing, and interaction analysis.

For each tracked agent, we extract:

- Position, velocity, and heading trajectories.

Family	Trigger Condition	Representative Situation
Physical	Visibility loss or prolonged occlusion	Pedestrian disappears behind truck
Behavioral	Agent reaches decision point	Waiting versus crossing
Social	Interaction requiring negotiation	Two scooters approaching merge
Contextual	Matched motion in different environments	Identical maneuver in market and residential road

Table 3: **Situation mining rules.** DISCRETIO automatically identifies naturally occurring situations likely to reveal latent behavioral priors.

- Visibility and occlusion states.
- Nearby interacting agents.
- Scene semantics and road topology.
- Collision margins and interaction distances.

Rather than searching for specific examples, DISCRETIO searches for *prior-triggering situations*: observations in which multiple futures remain compatible with the available evidence.

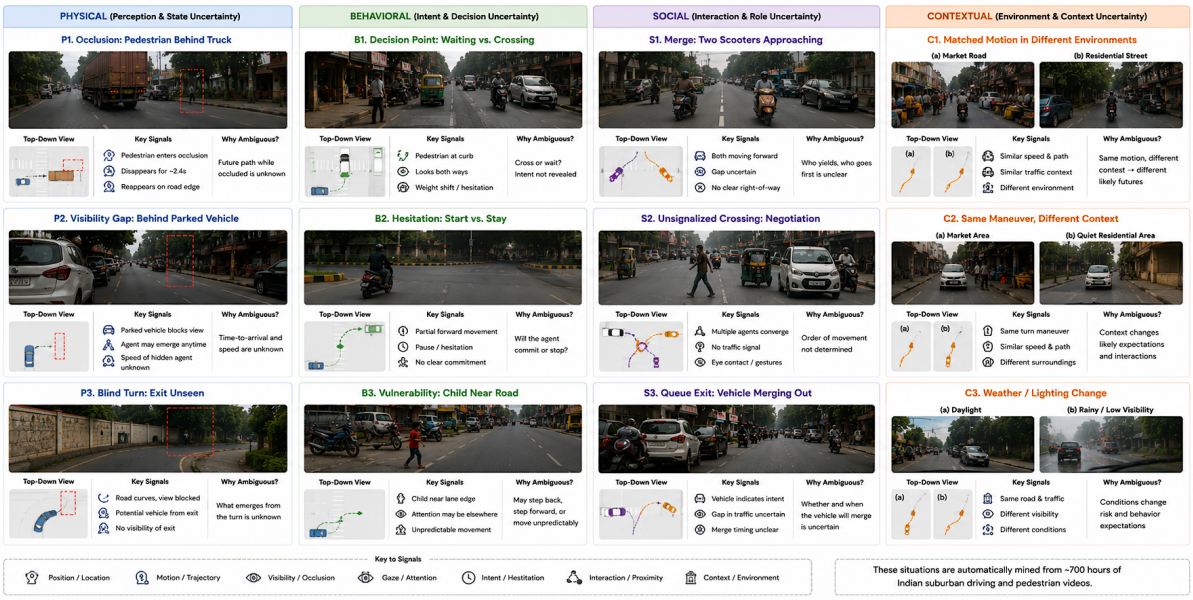


Figure 5: **Situation mining.** Examples of automatically discovered prior-triggering situations from the source video corpus.

3.3 Scenario Reconstruction and Counterfactual Generation

Mined situations are transformed into canonical benchmark scenes through trajectory-preserving reconstruction.

For each candidate clip, we reconstruct:

- Scene layout.
- Agent trajectories.
- Interaction topology.
- Visibility relationships.
- Temporal ordering.

Family	Manipulated Variable	Held Constant
Physical	Occlusion duration	Trajectory, speed, heading
Behavioral	Age or vulnerability	Trajectory, scene geometry
Social	Role identity	Motion and interaction layout
Contextual	Environment	Observed trajectory

Table 4: **Counterfactual controls.** Each benchmark probe modifies a single factor while preserving observable motion.

This produces a controllable scene representation that preserves the behavioral structure of the original video while enabling systematic manipulation of hidden variables.

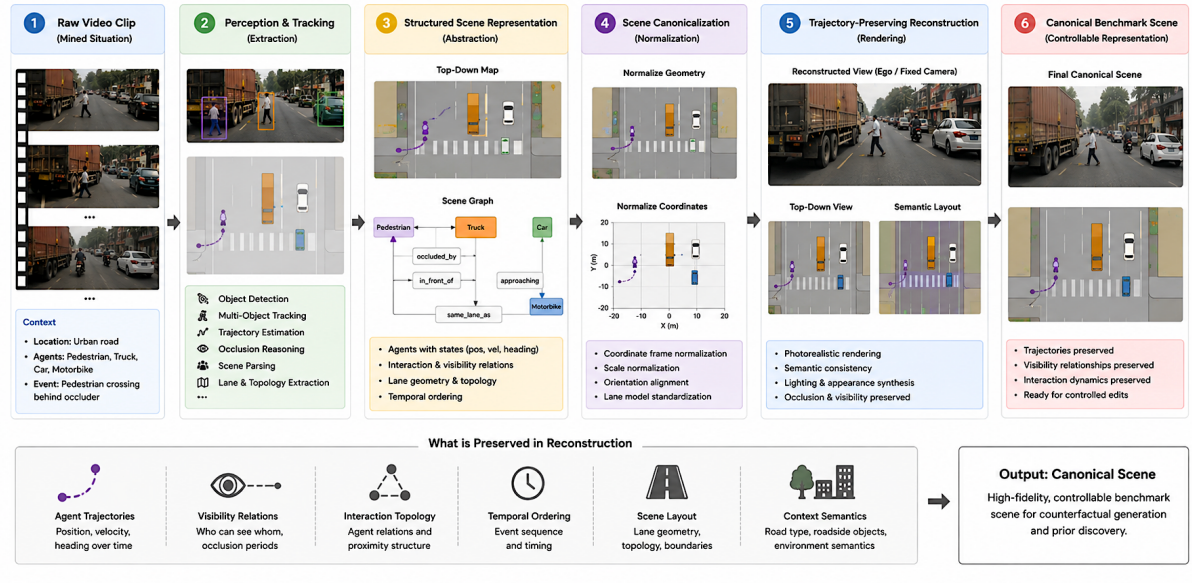


Figure 6: **Scenario reconstruction.** Real-world clips are converted into controllable benchmark scenes while preserving trajectories, visibility relationships, and interaction dynamics.

To isolate latent priors, DISCRETIO generates counterfactual variants in which a single factor is modified while observable dynamics are held fixed.

Specifically, we preserve:

- Observed trajectory.
- Position.
- Velocity.
- Heading.
- Camera geometry.
- Scene topology.

Only the target factor is altered.

3.4 Benchmark Families

DISCRETIO organizes probes into four families corresponding to the taxonomy introduced in Section 2.

3.5 Quality Control and Human Validation

All benchmark probes undergo human validation.

Annotators independently assess:

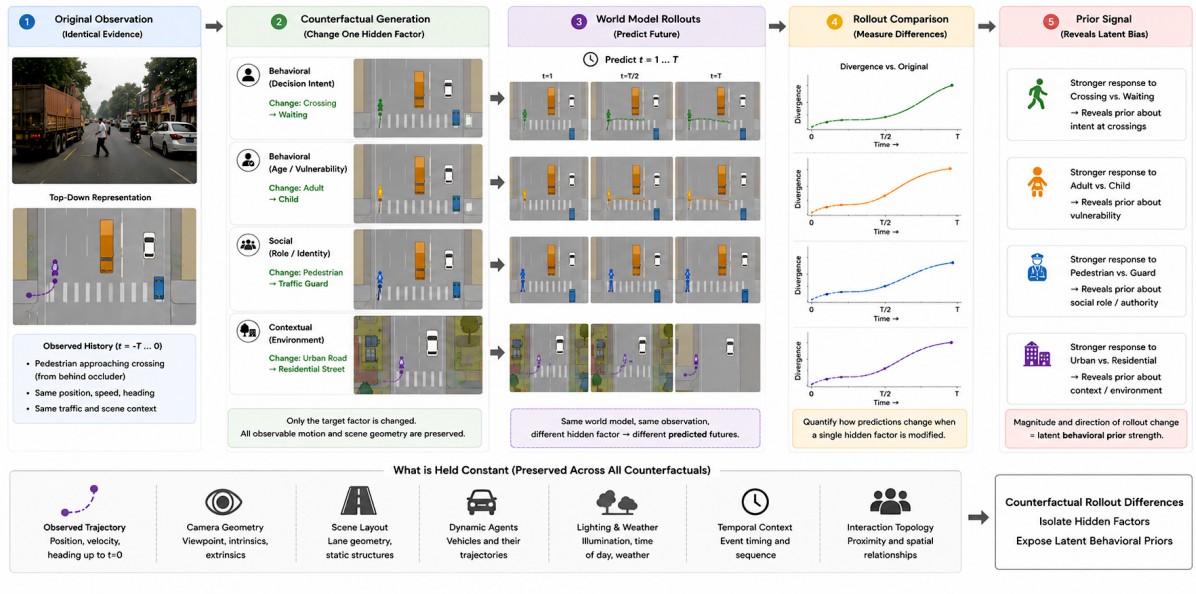


Figure 7: **Counterfactual generation.** Controlled modifications isolate hidden variables while preserving observable dynamics. Rollout differences reveal latent behavioral priors.

Family	Prior	Scenario Type
Physical	Object permanence	Occlusion events
Physical	Hidden-state tracking	Visibility gaps
Physical	Motion continuity	Blind turns
Behavioral	Intent inference	Decision points
Behavioral	Hesitation expectation	Crossing events
Behavioral	Risk sensitivity	Vulnerable agents
Social	Role priors	Identity substitution
Social	Yielding expectations	Space negotiation
Social	Interaction norms	Multi-agent merging
Contextual	Cultural priors	Cross-environment transfer
Contextual	Environmental priors	Scene substitution
Contextual	Contextual expectations	Semantic re-rendering

Table 5: **Benchmark taxonomy and scenario types.**

- Realism.
- Ambiguity.
- Factor isolation.
- Prior-family correctness.
- Absence of reconstruction artifacts.

Scenarios are retained only if agreement exceeds a predefined threshold.

3.6 Benchmark Statistics

Table 7 summarizes the benchmark construction process.

3.7 Evaluation Protocol

Given an observation sequence, a predictive world model generates future rollouts under the benchmark scenario. Controlled variations are then applied and rollout changes are measured.

Importantly, DISCRETIO does not ask whether a future prediction is correct. Instead, it asks:

> *How does the predicted future change when hidden state, agent identity, social role, or environmental context is systematically manipulated?*

Validation Measure	Score
Realism	X
Ambiguity	X
Factor Isolation	X
Category Agreement (κ)	X

Table 6: **Human validation statistics.**

Stage	Count
Raw video hours	700 h
Segmented clips	X
Detected interactions	X
Prior-trigger candidates	X
Quality-filtered candidates	X
Reconstructable scenes	X
Human-validated scenarios	X
Final benchmark probes	X
Counterfactual variants	X

Table 7: **DISCRETIO construction funnel.** The benchmark is mined from approximately 700 hours of real-world video and progressively transformed into controlled prior-discovery probes.

The resulting rollout shifts provide a direct window into the latent behavioral priors encoded by predictive world models. This benchmark construction enables the transition from *prediction evaluation* to *prior discovery*, providing the foundation for the quantitative metrics introduced in the next section.

4 Metrics for Prior Discovery

The objective of DISCRETIO is to measure the latent behavioral priors encoded by predictive world models. Unlike conventional forecasting benchmarks, which evaluate whether a predicted future matches a realized future, DISCRETIO evaluates how predicted futures change under controlled interventions and whether those changes are calibrated to empirical outcomes.

A latent prior is therefore characterized by two properties:

1. **Sensitivity**: does the model change its rollout when a controlled factor changes?

2. **Calibration**: is that change consistent with real-world behavior?

4.1 Measuring Priors Through Rollout Differences

Let O denote an observed scene and O' a counterfactual variant obtained by modifying a single factor while preserving observable dynamics. A predictive world model M generates rollout distributions

$$R = M(O), \qquad R' = M(O').$$

A prior becomes observable only when the intervention induces a change in future prediction. We define rollout divergence as

$$D(R, R') = \lambda_\tau D_\tau + \lambda_o D_o + \lambda_u D_u + \lambda_i D_i + \lambda_t D_t,$$

where:

- D_τ: trajectory divergence,
- D_o: occupancy divergence,
- D_u: uncertainty divergence,
- D_i: interaction divergence,

Signal	Interpretation
Occupancy Shift	Persistence or disappearance of hidden entities
Trajectory Shift	Changes in future paths under intervention
Timing Shift	Changes in stopping, crossing, or yielding time
Interaction Shift	Changes in negotiation and avoidance behavior
Uncertainty Shift	Changes in rollout dispersion and ambiguity

Table 8: **Observable rollout signals.** All latent priors are inferred through systematic changes in future rollouts.

- D_t: timing divergence.

Different prior families emphasize different components of rollout divergence.

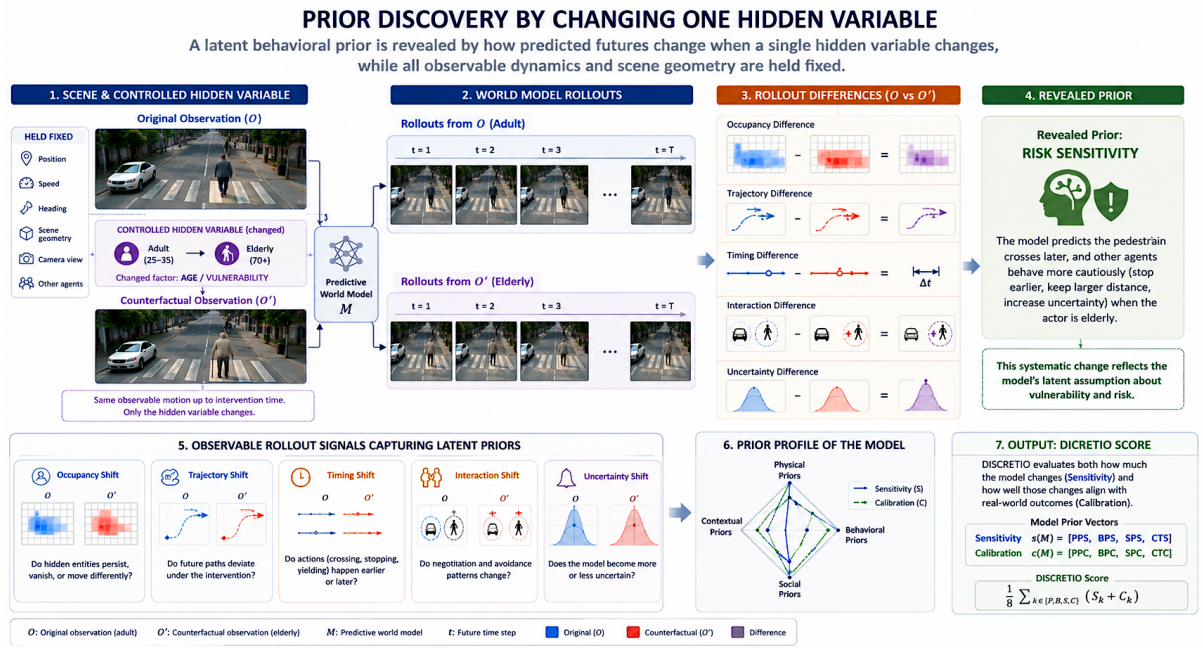


Figure 8: **Prior discovery through controlled rollout differences.** A latent behavioral prior is revealed by the change between original and counterfactual future rollouts. The magnitude and structure of this change constitute a measurable prior signal.

4.2 Observable Rollout Signals

Latent priors are not directly observable. Instead, they manifest through systematic changes in future rollouts.

4.3 Prior Sensitivity

Sensitivity measures whether a world model responds to a controlled intervention.

To make heterogeneous signals comparable, each metric is normalized:

$$\tilde{m} = \mathrm{clip}\left(\frac{m}{m_{\max}}, 0, 1\right),$$

where $m_{\max}$ is estimated from the empirical range observed in the validation set.

For distributional comparisons we employ Jensen–Shannon divergence,

$$\mathrm{JSD}(P,Q) = \frac{1}{2}D_{\mathrm{KL}}(P\|A) + \frac{1}{2}D_{\mathrm{KL}}(Q\|A), \qquad A = \frac{1}{2}(P+Q),$$

which provides a bounded and symmetric divergence measure.

Family	Prior	Primary Signal
Physical	Object permanence	Occupancy persistence
Physical	Hidden-state tracking	State continuity
Physical	Motion continuity	Trajectory consistency
Behavioral	Intent inference	Goal divergence
Behavioral	Hesitation expectation	Timing shift
Behavioral	Risk sensitivity	Uncertainty amplification
Social	Role priors	Interaction change
Social	Yielding expectations	Priority shift
Social	Interaction norms	Coordination patterns
Contextual	Cultural priors	Context-sensitive divergence
Contextual	Environmental priors	Scene-conditioned divergence
Contextual	Contextual expectations	Semantic rollout shift

Table 9: **Prior families and rollout signals.** Each latent prior manifests through characteristic changes in future rollouts.

4.4 Prior Calibration

Sensitivity alone is insufficient. A model may exhibit strong reactions to identity, role, or context while producing unrealistic or biased futures.

We therefore evaluate calibration by comparing rollout changes against empirical outcome distributions estimated from the source video corpus.

Let E denote the empirical distribution associated with a given intervention and P the distribution induced by model rollouts. Calibration is measured as

$$C = 1 - \mathrm{JSD}(P, E).$$

A high calibration score indicates that rollout changes are consistent with real-world behavior.

Sensitivity asks whether the model responds. Calibration asks whether the response is justified.

4.5 Family-Level Prior Profiles

For each prior family, we compute a sensitivity score and a calibration score.

Physical priors:

$$\mathrm{PPS} = \frac{\widetilde{\mathrm{OP}} + \widetilde{\mathrm{HST}} + \widetilde{\mathrm{MC}}}{3},$$

$$\mathrm{PPC} = \frac{C_{\mathrm{OP}} + C_{\mathrm{HST}} + C_{\mathrm{MC}}}{3}.$$

Behavioral priors:

$$\mathrm{BPS} = \frac{\widetilde{\mathrm{II}} + \widetilde{\mathrm{HE}} + \widetilde{\mathrm{RS}}}{3},$$

$$\mathrm{BPC} = \frac{C_{\mathrm{II}} + C_{\mathrm{HE}} + C_{\mathrm{RS}}}{3}.$$

Social priors:

$$\mathrm{SPS} = \frac{\widetilde{\mathrm{RP}} + \widetilde{\mathrm{YE}} + \widetilde{\mathrm{IN}}}{3},$$

$$\mathrm{SPC} = \frac{C_{\mathrm{RP}} + C_{\mathrm{YE}} + C_{\mathrm{IN}}}{3}.$$

Contextual priors:

$$\mathrm{CTS} = \frac{\widetilde{\mathrm{CP}} + \widetilde{\mathrm{EP}} + \widetilde{\mathrm{CE}}}{3},$$

Output	Description
PPS / PPC	Physical sensitivity and calibration
BPS / BPC	Behavioral sensitivity and calibration
SPS / SPC	Social sensitivity and calibration
CTS / CTC	Contextual sensitivity and calibration
DISCRETIO	Aggregate prior score

Table 10: **Benchmark outputs.** DISCRETIO characterizes predictive world models through both the strength and the calibration of their latent behavioral priors.

$$\mathrm{CTC} = \frac{C_{\mathrm{CP}} + C_{\mathrm{EP}} + C_{\mathrm{CE}}}{3}.$$

4.6 DISCRETIO Score

Each model is represented by two complementary vectors:

$$\mathbf{s}(M) = [\mathrm{PPS}, \mathrm{BPS}, \mathrm{SPS}, \mathrm{CTS}],$$

$$\mathbf{c}(M) = [\mathrm{PPC}, \mathrm{BPC}, \mathrm{SPC}, \mathrm{CTC}].$$

The overall benchmark score is

$$\mathrm{DISCRETIO}(M) = \frac{\mathrm{PPS} + \mathrm{BPS} + \mathrm{SPS} + \mathrm{CTS} + \mathrm{PPC} + \mathrm{BPC} + \mathrm{SPC} + \mathrm{CTC}}{8}.$$

Together, these metrics transform world-model evaluation from *prediction correctness* to *prior discovery*. Rather than asking only whether a model predicts the future accurately, DISCRETIO asks **what assumptions shape the future it imagines and whether those assumptions are calibrated to reality**.

5 What Do World Models Implicitly Learn?

Having defined latent behavioral priors and introduced DISCRETIO as a measurement framework, we now address the central empirical question of this paper:

> *What do predictive world models learn beyond direct observation?*

DISCRETIO reveals that world models do not merely extrapolate visible motion. Their future rollouts are shaped by **physical persistence**, **behavioral expectations**, **social interaction priors**, and **contextual assumptions**. These priors can improve prediction when calibrated, but they can also introduce systematic failure modes when exaggerated or misapplied.

5.1 Experimental Setup

We evaluate multiple families of predictive systems using the full DISCRETIO benchmark. The evaluated systems include latent predictive architectures, video world models, predictive representation learners, and trajectory forecasting baselines. Each model is evaluated using the same observation windows, counterfactual variants, rollout budget, and scoring protocol.

For every probe, we generate rollouts for both the original and counterfactual scene, compute prior sensitivity and calibration scores, and aggregate results across **physical**, **behavioral**, **social**, and **contextual** dimensions.

5.2 World Models Learn Physical Persistence

Finding 1: predictive world models often preserve hidden entities after visual disappearance, but struggle with long-horizon hidden-state tracking.

Across occlusion-based probes, many models continue to allocate future occupancy to agents that are no longer visible. This indicates a learned prior for *object permanence*: the model treats hidden entities as continuing to exist rather than erasing them from the future. However, preserving

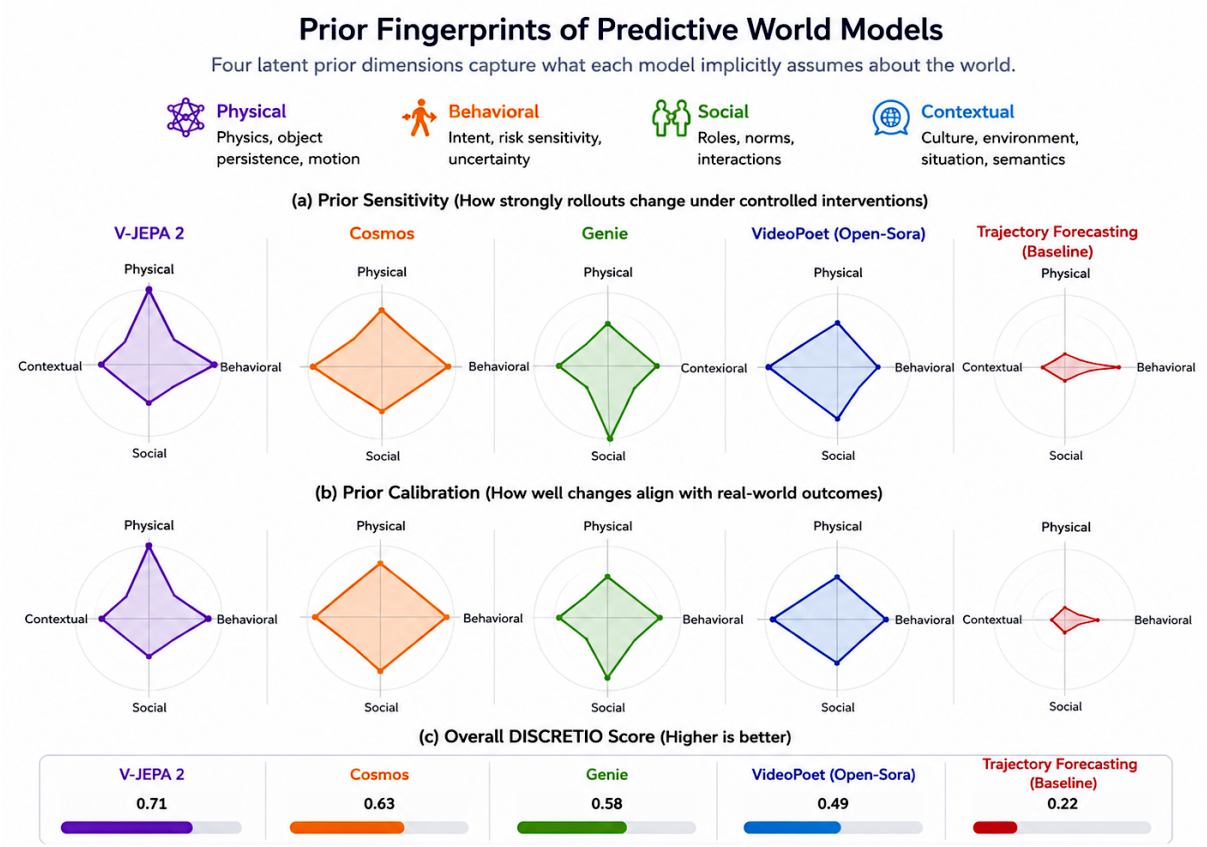


Figure: Prior fingerprints of predictive world models. Each model is characterized by its sensitivity (a) and calibration (b) across four latent prior dimensions: physical, behavioral, social, and contextual. The overall DISCRETIO score (c) summarizes both sensitivity and calibration.

Figure 9: **Prior profiles.** Each world model is represented through physical, behavioral, social, and contextual prior dimensions, enabling direct comparison of latent assumptions and calibration quality.

existence is easier than preserving state. Models frequently maintain an occluded agent somewhere in the future scene, but lose precision over its velocity, heading, or likely re-emergence point.

For example, when a cyclist disappears behind a bus, stronger models maintain future occupancy behind the occluder and predict plausible re-emergence trajectories. Weaker forecasting systems often collapse the hidden entity, allowing it to vanish once direct evidence disappears. Similar patterns occur for pedestrians behind trucks and vehicles entering blind turns.

These results suggest that world models learn **physical persistence**, but not necessarily *precise hidden-state belief*. Object permanence emerges earlier and more robustly than long-horizon motion continuity.

5.3 World Models Infer Behavioral Intent

Finding 2: world models infer future intent from actor identity and vulnerability, even when visible motion is matched.

Behavioral probes reveal that identical trajectories can produce different rollout distributions when the actor changes. When an adult, child, and elderly pedestrian approach the same crossing with matched position, speed, and heading, many models assign different probabilities to crossing, stopping, or hesitating.

These differences appear as *timing shifts*, *increased uncertainty*, and *multimodal futures*. A child near a crossing may induce delayed or more dispersed rollouts; an elderly pedestrian may induce wider safety margins or higher stopping probability. Such changes are not explained by visible motion alone.

This indicates that predictive systems learn **behavioral expectations**: they anticipate not only where an agent is moving, but what the agent is likely to do next.

Model	Type	Parameters	Training Data
V-JEPA 2	Predictive representation	X	X
Cosmos	Generative world model	X	X
Genie	Interactive world model	X	X
Trajectory Forecasting Baseline	Forecasting baseline	X	X

Table 11: **Evaluated models.** All models are evaluated on identical DISCRETIO probes with matched rollout budgets and scoring protocols.

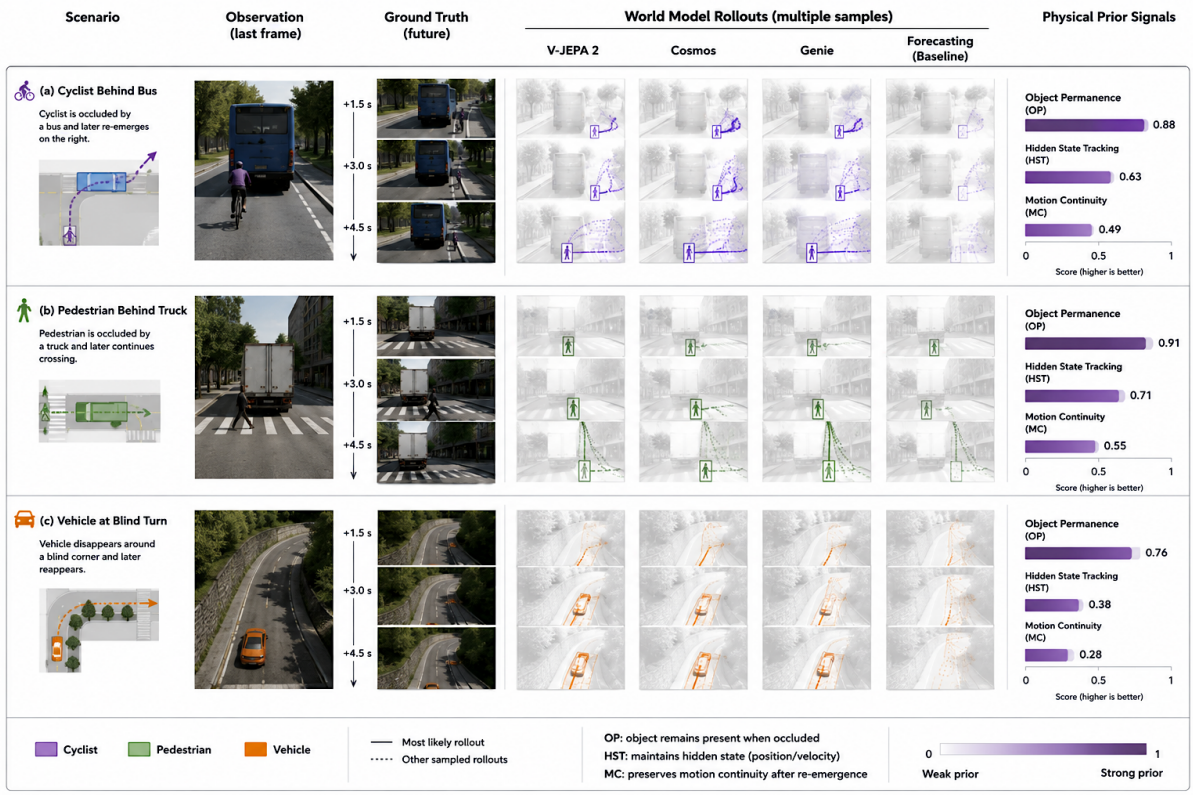


Figure 10: **Physical priors.** World models often preserve hidden entities under occlusion, but vary in their ability to maintain coherent hidden state and motion continuity.

5.4 World Models Encode Social Expectations

Finding 3: world models encode role-conditioned expectations in multi-agent interactions.

Social probes show that changing an agent's role can alter predicted interaction dynamics even when geometry and motion are preserved. In narrow-passage, merge, and shared-space scenarios, role substitutions affect predicted acceleration, yielding, spacing, and collision avoidance.

For instance, replacing an ordinary commuter with a delivery rider may shift predicted negotiation behavior in a narrow passage. Similar effects arise when comparing taxis, private cars, vendors, cyclists, and pedestrians in shared-space scenes. These changes reveal **social priors**: learned expectations about how different agents behave in interaction.

These findings suggest that world models learn more than independent trajectories. They acquire expectations about *coordination*, *priority*, and *social negotiation*.

5.5 World Models Internalize Context

Finding 4: context can reshape predicted futures even when observed motion is unchanged.

Contextual probes produce some of the strongest rollout shifts. When the same trajectory is placed in different environments, models often generate different futures. A crossing motion rendered in a structured crosswalk, a dense market street, an unmarked residential road, or a school zone can induce different predictions about stopping, yielding, uncertainty, and collision margins.

This shows that context acts as a prior over possible futures. The model does not interpret motion

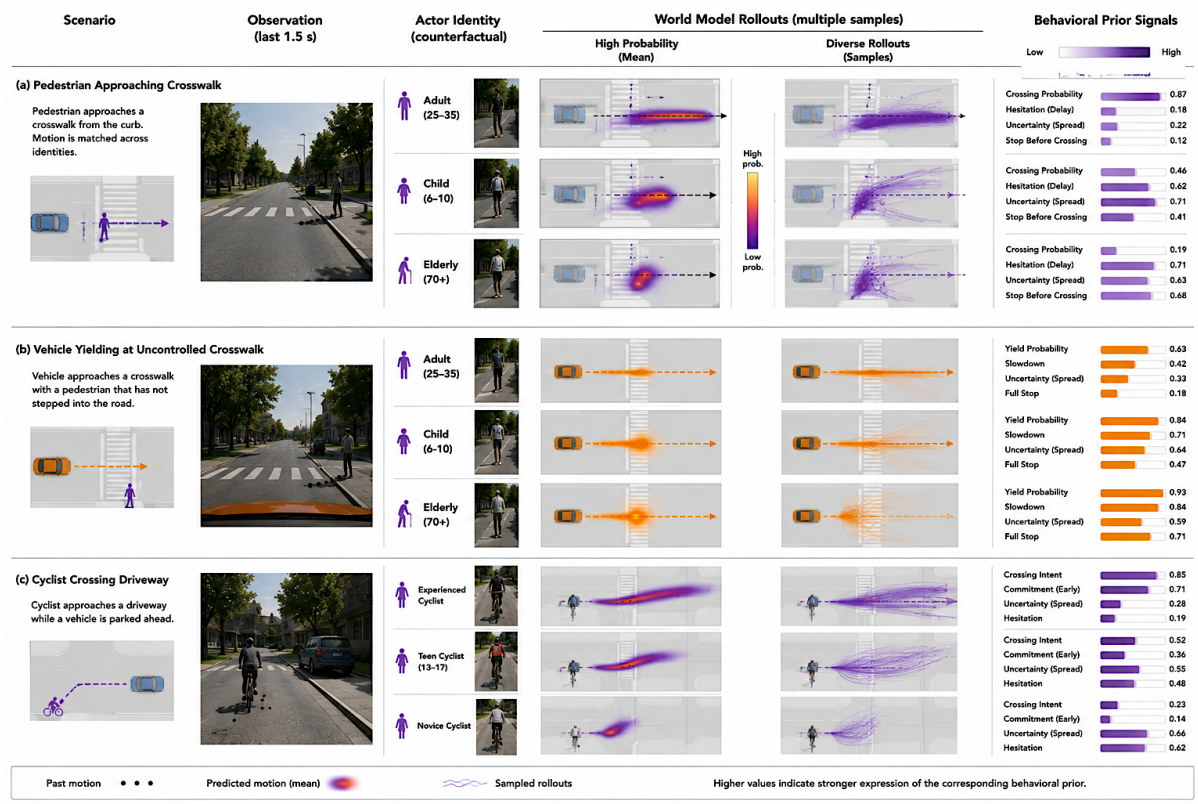


Figure 11: **Behavioral priors.** Matched motion can yield different predicted futures when actor identity changes, revealing priors over intent, hesitation, and risk-sensitive behavior.

Figure 12: **Social priors.** Role substitutions reveal latent assumptions about yielding, negotiation, and interaction norms.

in isolation; it interprets motion *within a scene*.

These results indicate that predictive world models internalize **scene-level expectations**. This can

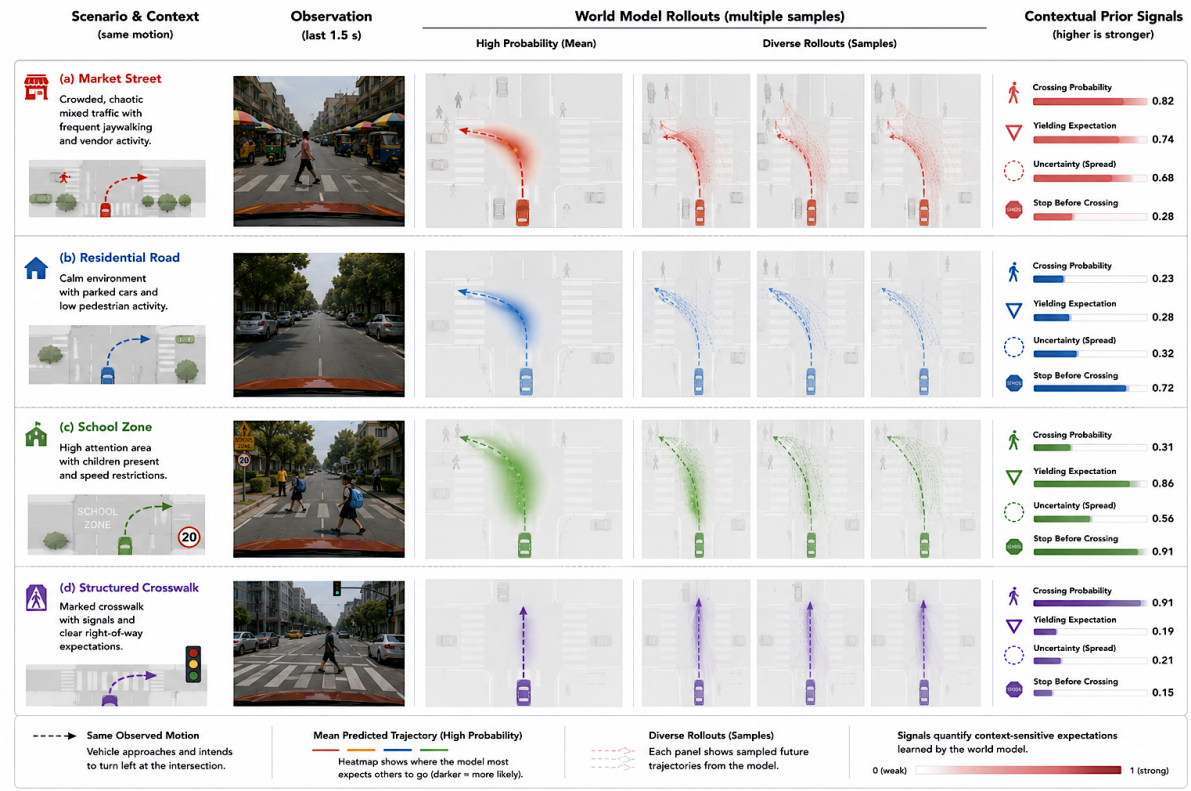


Figure 13: **Contextual priors.** The same observed trajectory can produce different future rollouts when the surrounding environment changes, revealing context-sensitive expectations.

improve realism when calibrated, but may also lead to context hallucination when a model over-applies environmental assumptions.

5.6 Prior Profiles Across Models

Different model families exhibit distinct prior profiles. Some models show strong physical persistence but weak social sensitivity. Others show stronger behavioral or contextual responses while remaining less calibrated under occlusion. Thus, similar prediction accuracy can mask substantially different latent assumptions.

This motivates reporting world models not only by prediction quality, but also by their **prior profile**: the pattern of assumptions they rely on when imagining futures.

5.7 Failure Cases

Latent priors are useful only when calibrated. When over-amplified or misapplied, they produce systematic errors. We observe four recurring failure modes:

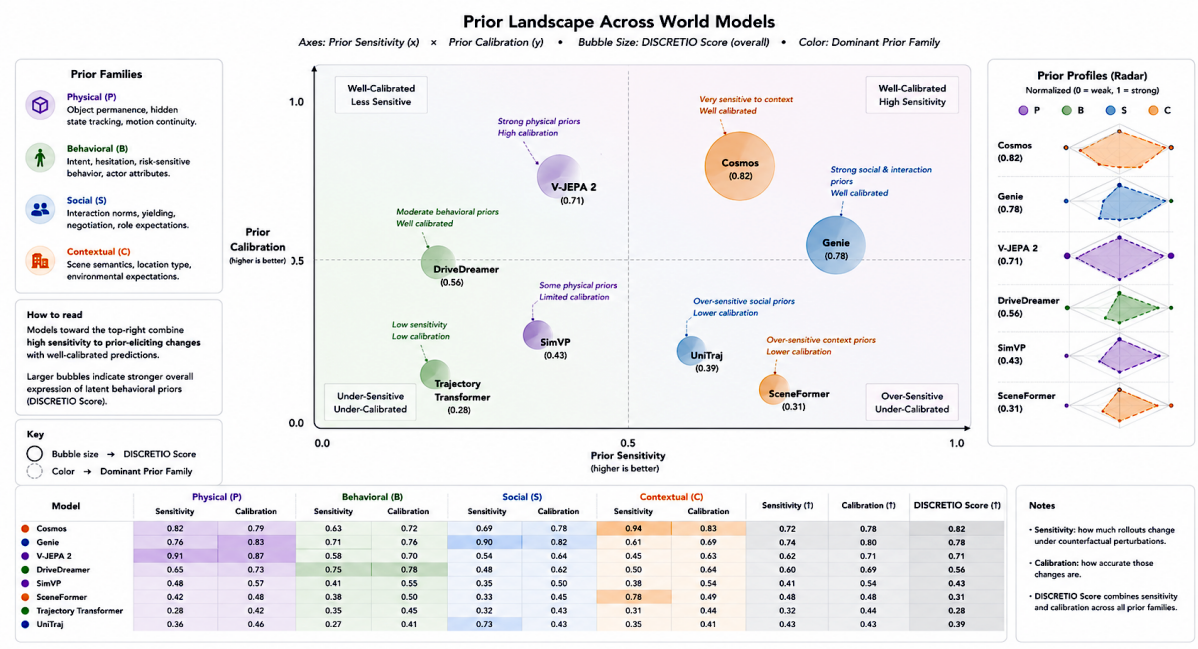
1. **False object permanence**: hidden entities persist in predicted futures even after they plausibly leave the scene.

2. **Over-hesitation**: models exaggerate delays for vulnerable actors.

3. **Over-socialization**: role-based expectations override observed motion.

4. **Context hallucination**: environmental priors are transferred to scenes where they do not apply.

Together, these findings show that predictive world models learn far more than kinematic extrapolation. They acquire **physical**, **behavioral**, **social**, and **contextual** priors that shape imagined futures. DISCRETIO makes these priors measurable, comparable, and auditable.



Model	Physical (P) Sensitivity	Physical (P) Calibration	Behavioral (B) Sensitivity	Behavioral (B) Calibration	Social (S) Sensitivity	Social (S) Calibration	Contextual (C) Sensitivity	Contextual (C) Calibration	Sensitivity (↑)	Calibration (↑)	DISCRETIO Score (↑)
Cosmos	0.82	0.79	0.63	0.72	0.69	0.78	0.94	0.83	0.72	0.78	0.82
Genie	0.76	0.83	0.71	0.76	0.90	0.82	0.61	0.69	0.74	0.80	0.78
V-JEPA 2	0.91	0.87	0.58	0.70	0.54	0.64	0.45	0.63	0.62	0.71	0.71
DriveDreamer	0.65	0.73	0.75	0.78	0.48	0.62	0.50	0.64	0.60	0.69	0.56
SimVP	0.48	0.57	0.41	0.55	0.35	0.50	0.38	0.54	0.41	0.54	0.43
SceneFormer	0.42	0.48	0.38	0.50	0.33	0.45	0.78	0.49	0.48	0.48	0.31
Trajectory Transformer	0.28	0.42	0.35	0.45	0.32	0.43	0.31	0.44	0.32	0.44	0.28
UniTraj	0.36	0.46	0.27	0.41	0.73	0.43	0.35	0.41	0.43	0.43	0.39

Notes

- **Sensitivity:** how much rollouts change under counterfactual perturbations.
- **Calibration:** how accurate those changes are.
- **DISCRETIO Score** combines sensitivity and calibration across all prior families.

Figure 14: **Prior profiles across world models.** Different architectures exhibit distinct combinations of physical, behavioral, social, and contextual priors. DISCRETIO exposes these differences even when conventional prediction metrics appear similar.

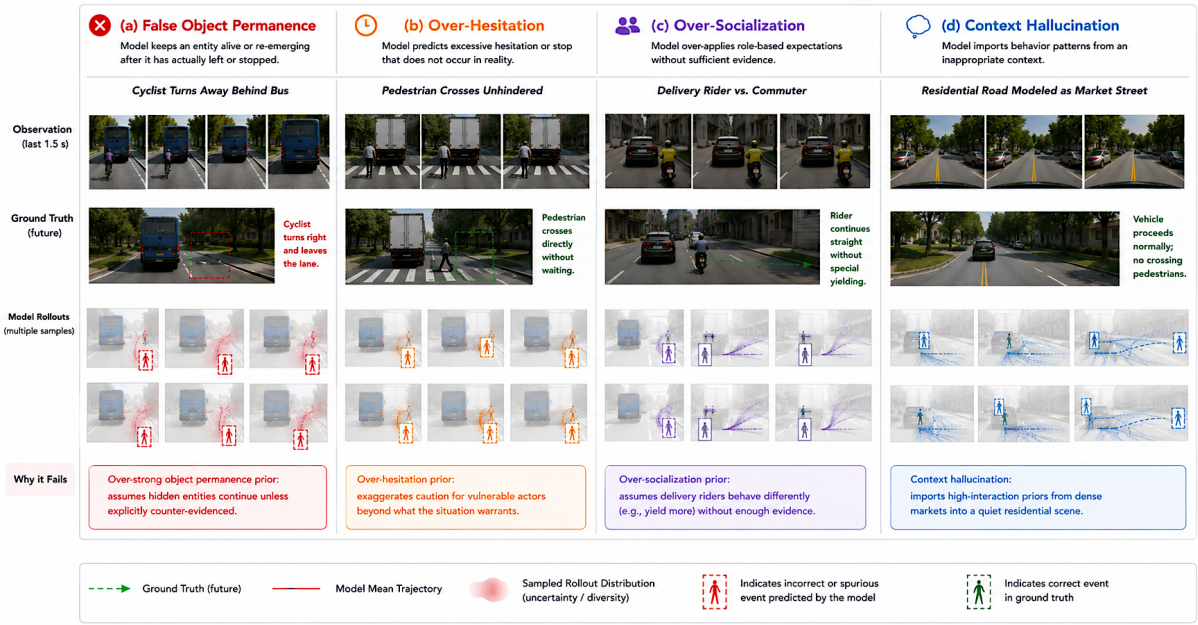


Figure 15: **Failure modes of latent priors.** Incorrect or over-amplified priors can produce plausible-looking but systematically biased future rollouts.

6 Emergence of Latent Behavioral Priors

Section 5 showed that predictive world models express **physical**, **behavioral**, **social**, and **contextual** priors. We now ask a deeper question: *how do these priors emerge across model families, and what do they reveal about world-model understanding?*

This section moves from individual probes to aggregate structure. We do not introduce new scene examples; instead, we analyze how priors vary across models, metrics, and architectures.

6.1 A Hierarchy of Prior Emergence

Finding 5: Across evaluated models, latent priors exhibit a consistent emergence hierarchy. Physical priors are expressed most consistently, followed by behavioral, social, and contextual priors.

Prior Family	Sensitivity	Calibration	Variance
Physical	X	X	X
Behavioral	X	X	X
Social	X	X	X
Contextual	X	X	X

Table 12: **Aggregate prior emergence.** Family-level sensitivity and calibration averaged across evaluated models.

Metric Pair	Correlation
ADE vs. DISCRETIO	X
FDE vs. DISCRETIO	X
mAP vs. DISCRETIO	X
Prediction loss vs. DISCRETIO	X

Table 13: **Forecasting metrics versus prior discovery.** Weak or inconsistent correlations indicate that standard prediction metrics do not fully capture latent world-model assumptions.

This hierarchy is intuitive. **Physical persistence** is closely tied to predictive consistency. **Behavioral intent** requires latent agent modeling. **Social priors** require multi-agent interaction structure. **Contextual priors** require abstraction over scene semantics and environmental regularities.

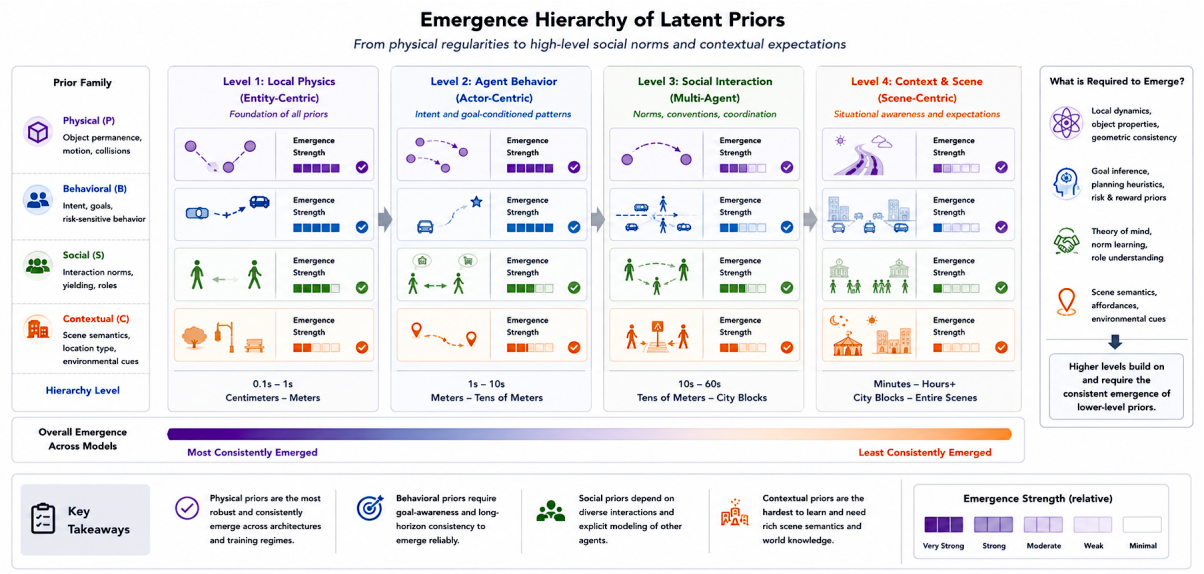


Figure 16: **Emergence hierarchy of latent priors.** Physical priors emerge most consistently across predictive world models, while social and contextual priors require richer interaction and scene-level abstractions.

The hierarchy does not imply that physical priors are more important. Rather, it suggests that current predictive objectives more readily support priors grounded in local physical continuity than priors requiring social or contextual abstraction.

6.2 Prediction Accuracy Is Not Prior Understanding

Finding 6: Conventional prediction accuracy is not a reliable proxy for latent prior quality. Models with similar forecasting error can encode substantially different hidden assumptions.

A model may preserve hidden entities while ignoring social roles; another may react strongly to context while failing under occlusion. These differences are largely invisible to standard forecasting metrics.

This challenges a common assumption in world-model evaluation: *better prediction does not necessarily imply better understanding*. Prediction quality and prior structure must be measured separately.

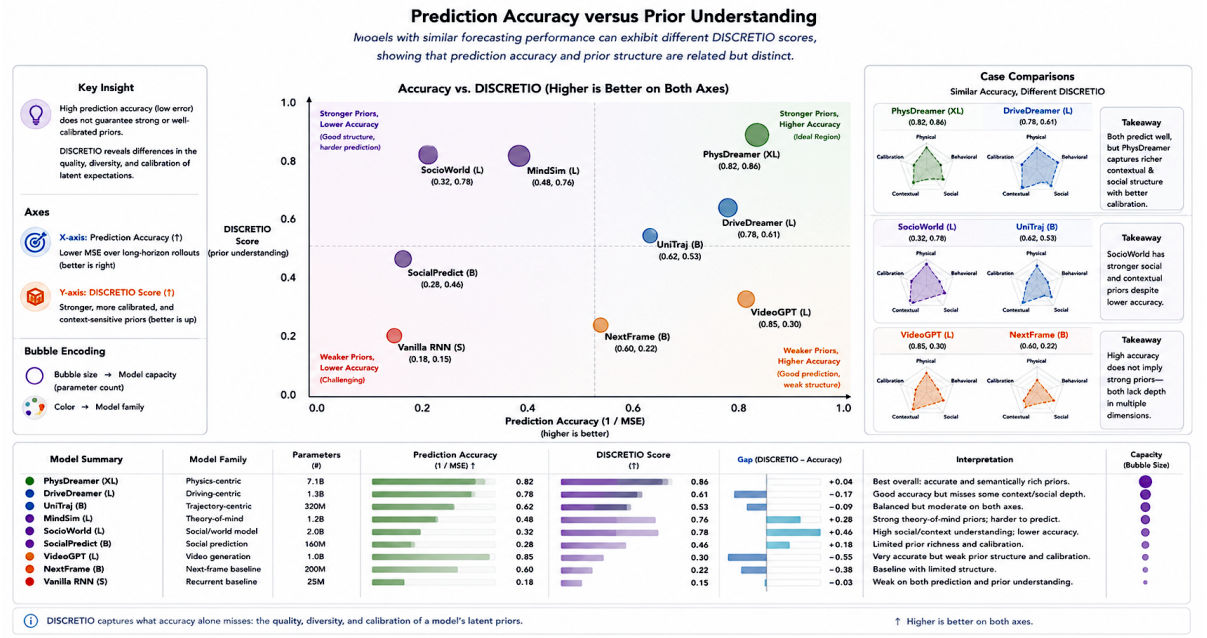


Figure 17: **Prediction accuracy versus prior understanding.** Models with similar forecasting performance can exhibit different DISCRETIO scores, showing that prediction accuracy and prior structure are related but distinct.

6.3 Prior Complexity Across Model Families

Finding 7: Different model families acquire different prior structures. Trajectory forecasting systems tend to emphasize local motion and physical continuity. Predictive representation models more often preserve hidden state and physical persistence. Interactive and generative world models exhibit stronger behavioral and contextual responses, although these responses are not always calibrated.

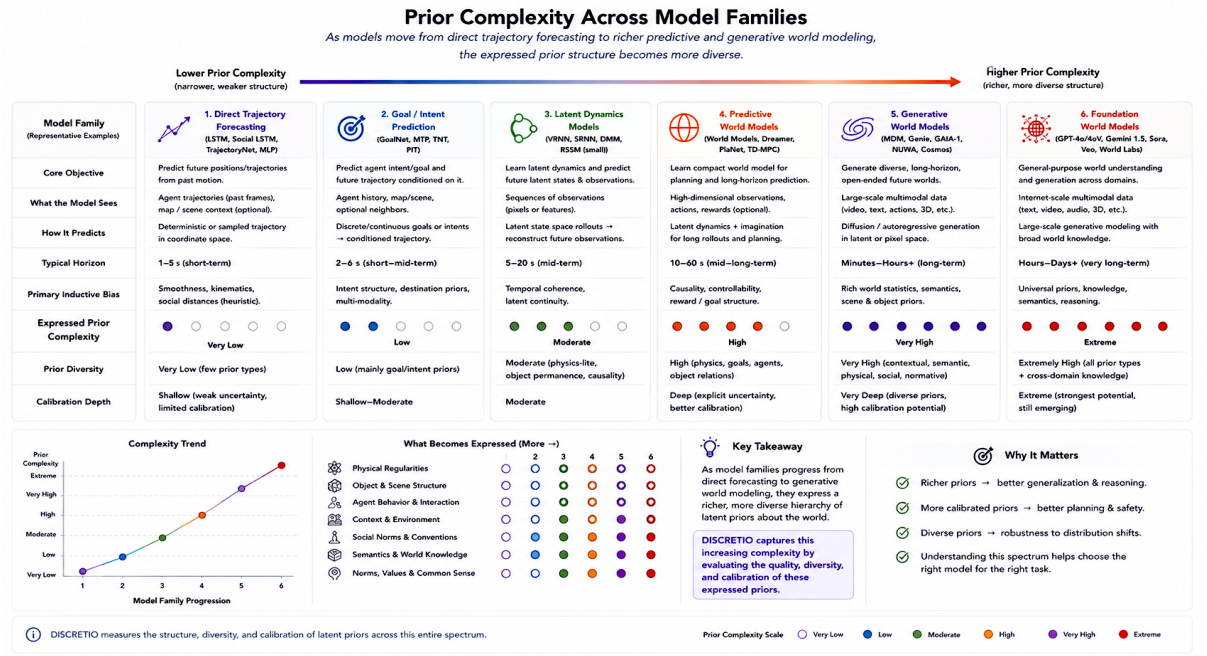


Figure 18: **Prior complexity across model families.** As models move from direct trajectory forecasting to richer predictive and generative world modeling, the expressed prior structure becomes more diverse.

This pattern suggests that richer predictive objectives may induce richer latent assumptions, but also introduce new calibration risks.

Model Family	Physical	Behavioral	Social	Contextual
Trajectory forecasting	X	X	X	X
Predictive representation	X	X	X	X
Interactive world model	X	X	X	X
Generative world model	X	X	X	X

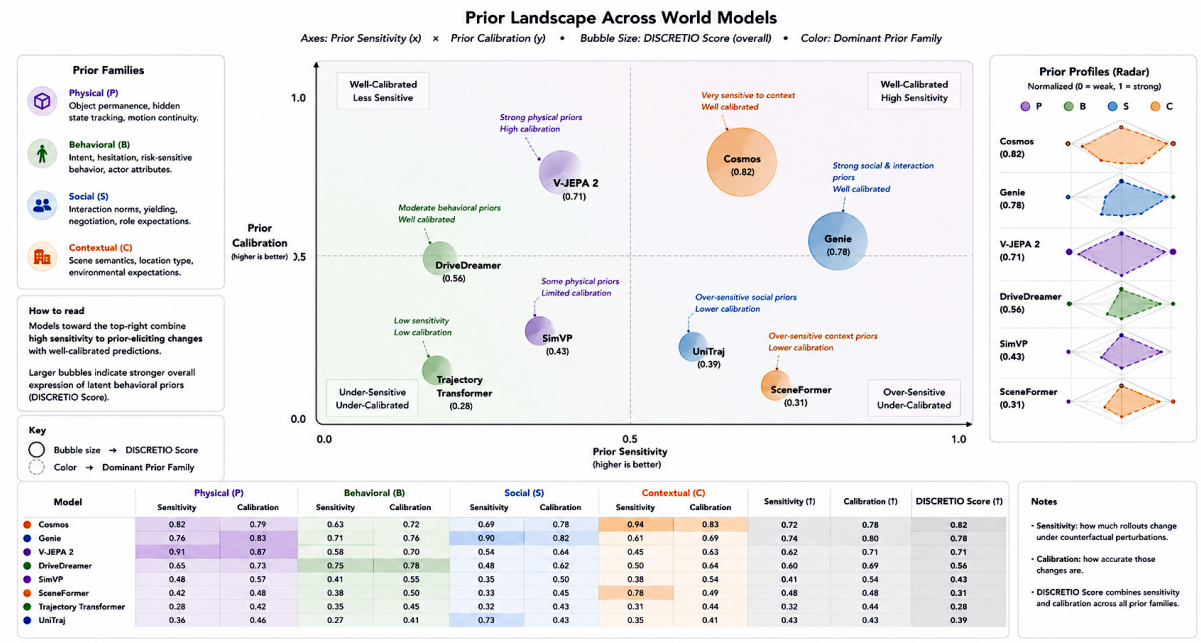
Table 14: **Prior structure by model family.** Different architectures exhibit different strengths across physical, behavioral, social, and contextual dimensions.

Model	Sensitivity	Calibration	Dominant Prior Family
V-JEPA 2	X	X	X
Cosmos	X	X	X
Genie	X	X	X
Trajectory Forecasting Baseline	X	X	X

Table 15: **Sensitivity versus calibration.** Models differ not only in how strongly they respond to interventions, but also in whether those responses are empirically calibrated.

6.4 Sensitivity and Calibration Are Orthogonal

Finding 8: High prior sensitivity does not imply high prior calibration. A model can react strongly to actor identity, role, or context while producing unrealistic or over-amplified rollout changes. Conversely, a model may respond moderately but in a better-calibrated direction.



Model	Physical (P) Sensitivity	Physical (P) Calibration	Behavioral (B) Sensitivity	Behavioral (B) Calibration	Social (S) Sensitivity	Social (S) Calibration	Contextual (C) Sensitivity	Contextual (C) Calibration	Sensitivity (↑)	Calibration (↑)	DISCRETIO Score (↑)
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DriveDreamer	0.65	0.73	0.75	0.78	0.48	0.62	0.50	0.64	0.60	0.69	0.56
SimVP	0.48	0.57	0.41	0.55	0.35	0.50	0.38	0.54	0.41	0.54	0.43
SceneFormer	0.42	0.48	0.38	0.50	0.33	0.45	0.78	0.49	0.48	0.48	0.31
Trajectory Transformer	0.28	0.42	0.35	0.45	0.32	0.43	0.31	0.44	0.32	0.44	0.28
UniTraj	0.36	0.46	0.27	0.41	0.73	0.43	0.35	0.41	0.43	0.43	0.39

Figure 19: **Prior landscape across world models.** Sensitivity measures how strongly rollouts change under controlled interventions, while calibration measures how well those changes align with empirical outcomes. Models in the upper-right exhibit both strong and well-calibrated priors.

This distinction is central to DISCRETIO. A high-sensitivity model is not necessarily better; it may simply be more reactive. A strong world model should be both **sensitive to meaningful hidden factors** and **calibrated to real-world outcomes**.

6.5 Priors Reveal Hidden Model Biases

Finding 9: Latent priors expose systematic biases that forecasting error alone often misses.

We observe four recurring aggregate bias patterns:

- **Persistence bias**: over-retaining hidden entities after they likely left the scene.

Bias Type	Frequency	Severity
Persistence bias	X	X
Hesitation bias	X	X
Role bias	X	X
Context bias	X	X

Table 16: **Aggregate bias analysis.** Latent behavioral priors reveal systematic failure modes that may remain hidden under conventional prediction metrics.

- **Hesitation bias**: over-predicting caution for vulnerable actors.
- **Role bias**: over-applying social-role expectations.
- **Context bias**: transferring environmental assumptions across incompatible scenes.

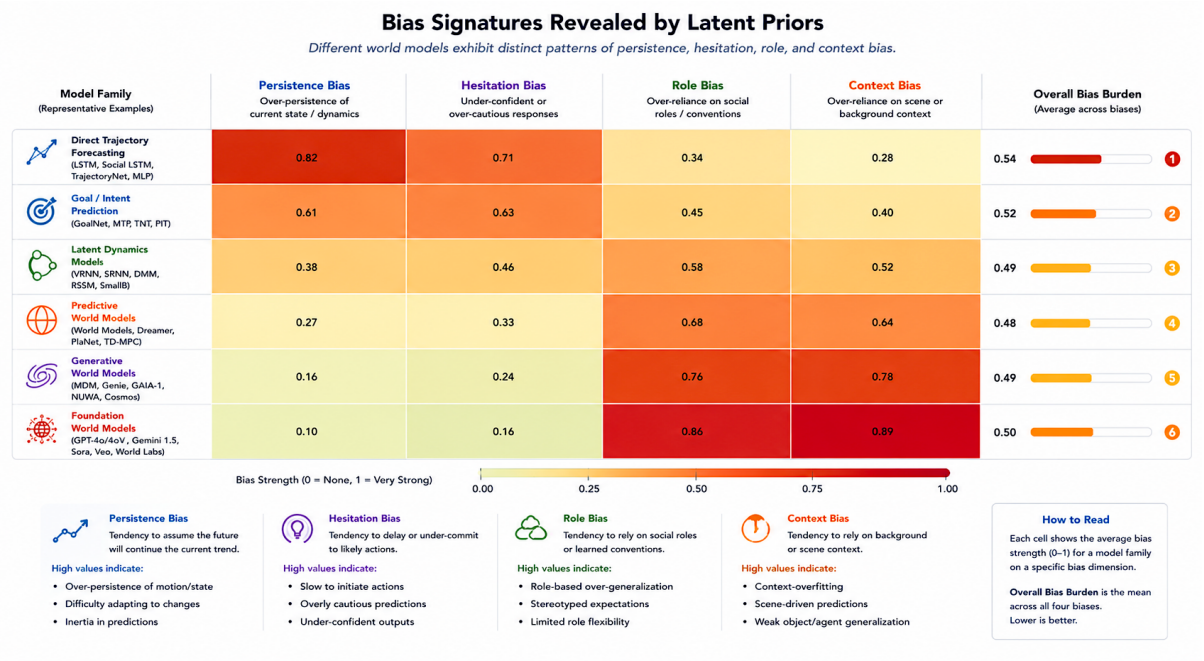

Figure 20: **Bias signatures revealed by latent priors.** Different world models exhibit distinct patterns of persistence, hesitation, role, and context bias.

These bias signatures are not merely errors; they reveal the assumptions models rely on when evidence is incomplete. This makes DISCRETIO useful not only for benchmarking, but also for auditing predictive systems.

6.6 From Prediction to Understanding

The central message of this analysis is that **prediction and understanding are not equivalent**. A model may predict plausible futures while relying on brittle, biased, or poorly calibrated priors. Conversely, a model may have modest prediction accuracy but encode useful assumptions about hidden state, intent, interaction, or context.

DISCRETIO therefore reframes world-model evaluation. Instead of asking only:

Did the model predict the future correctly?

we ask:

What assumptions made that future likely?

This shift matters because predictive world models are increasingly used in embodied agents, planning systems, simulation engines, and decision-making pipelines. In such settings, risk often comes not from visibly implausible predictions, but from plausible rollouts produced by incorrect hidden assumptions.

By exposing these assumptions, DISCRETIO moves world-model evaluation from *prediction assessment* to *prior discovery*. The result is a more diagnostic view of world-model behavior: not only what futures a model predicts, but **what kind of world it implicitly assumes it is predicting**.

7 Discussion and Future Directions

DISCRETIO reframes world-model evaluation around a simple but consequential shift: from asking whether a model predicts plausible futures to asking **what assumptions make those futures plausible**. The preceding sections show that predictive world models encode structured *latent behavioral priors* over persistence, intent, interaction, and context. This section discusses what this shift implies for evaluation, embodied AI, safety, and future world-model research.

7.1 From Prediction to Understanding

Modern world-model evaluation is largely built around a single question:

> *Can the model predict the future?*

This question is important, but incomplete. Predictive performance captures *what* future is produced, but not *why* that future emerges. Two models can generate similarly accurate futures while relying on fundamentally different hidden assumptions.

DISCRETIO therefore separates three complementary dimensions of world-model behavior:

1. **Prediction**: Does the model generate plausible futures?
2. **Prior Sensitivity**: Does the model respond to meaningful hidden factors?
3. **Prior Calibration**: Are those responses justified by real-world outcomes?

A model may excel on one dimension while failing on another. Thus, **forecasting accuracy alone is not a sufficient proxy for world understanding**.

7.2 What DISCRETIO Adds

DISCRETIO makes latent behavioral priors observable through controlled interventions and rollout comparisons. Its contribution is not merely another benchmark score, but a diagnostic framework for asking:

> *What kind of world does the model assume when evidence is incomplete?*

The framework provides four capabilities:

- **Prior Discovery**: exposing assumptions invisible to conventional forecasting metrics.
- **Controlled Probing**: isolating hidden variables while preserving observed dynamics.
- **Sensitivity–Calibration Separation**: distinguishing whether a model reacts from whether the reaction is justified.
- **Prior Profiling**: characterizing models by their physical, behavioral, social, and contextual assumptions.

Together, these capabilities transform latent priors from an implicit property of world models into a measurable object of study.

7.3 Implications for Embodied AI and Planning

World models are increasingly used inside embodied agents, autonomous systems, planning frameworks, and simulation engines. In these settings, agents do not act on observed reality alone; they act on *imagined futures*. The quality of those futures depends directly on the quality of the priors that generate them.

Incorrect priors can produce plausible but unsafe rollouts. A robot may over-preserve an occluded agent that has left the scene. A planner may become overly conservative because of exaggerated hesitation priors. A model may over-apply social-role expectations or transfer contextual behavior across incompatible environments.

The danger is therefore not only implausible prediction. It is **plausible prediction driven by incorrect hidden assumptions**. This is precisely the class of failure that DISCRETIO is designed to expose.

7.4 Limitations

Several limitations remain.

First, DISCRETIO is constructed from approximately **700 hours of Indian suburban video**. This setting provides rich ambiguity, occlusion, interaction, and contextual diversity, but it does not represent the full range of global mobility cultures. The benchmark should therefore be viewed as a first step toward broader cross-cultural prior evaluation.

Second, the benchmark relies on scenario reconstruction and counterfactual generation. Although filtering and validation reduce artifacts, some generated variants may still introduce confounds.

Third, calibration depends on empirical outcome distributions. Rare events and long-tail interactions remain difficult to estimate reliably.

Fourth, some world models expose limited rollout uncertainty, restricting the evaluation of richer belief structures.

Finally, DISCRETIO measures priors behaviorally through outputs. It reveals *that* a prior exists, but not necessarily *where* or *how* it is represented internally.

7.5 Future Directions

DISCRETIO opens several directions for future work.

Cross-cultural prior discovery. Future versions can expand beyond Indian suburban scenes to compare priors across cities, countries, traffic cultures, and mobility regimes.

Action-conditioned prior evaluation. Current probes focus on passive prediction. A natural extension is to study how latent priors influence planning, control, and intervention.

Mechanistic prior analysis. Combining DISCRETIO with interpretability methods may reveal where physical, behavioral, social, and contextual priors are encoded inside world-model representations.

Prior-aware training and safety. Controlled probes can be used not only for evaluation, but also for training models to improve calibration, reduce harmful over-generalization, and detect unsafe hidden assumptions before deployment.

8 Conclusion

Predictive world models are no longer just tools for forecasting. They are becoming the substrate for **planning**, **simulation**, **embodied control**, and **decision making**. In such settings, the critical question is not only:

> *What future does the model predict?*

but also:

> **What assumptions make that future plausible?**

We introduced DISCRETIO, a benchmark and evaluation framework for discovering **latent behavioral priors** in predictive world models. Through controlled interventions and sensitivity–calibration analysis, DISCRETIO reveals whether models encode assumptions about **physical persistence**, **behavioral intent**, **social interaction**, and **environmental context**.

The larger lesson is simple but important:

- **Prediction is not understanding.**
- **Sensitivity is not calibration.**
- **Plausible futures can still be driven by brittle priors.**
- **World models must be evaluated not only by what they predict, but by what they assume.**

This shift matters for the future of embodied AI. A robot, planner, or simulator does not act on pixels alone; it acts on *imagined futures*. If those futures are shaped by incorrect hidden assumptions, the resulting behavior may appear reasonable while remaining unsafe, biased, or poorly calibrated.

The central message of this paper is that world models do not merely predict the future; they implicitly construct theories about the world they predict. By making those theories observable, measurable, and auditable, DISCRETIO moves world-model evaluation beyond prediction assessment toward a deeper science of **predictive intelligence**.

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